An Automated Technique for Image Noise Identification Using a Simple Pattern Classification Approach

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Abstract - A simple pattern classification based noise identification method is proposed in this paper. The key idea involves isolation of some representative noise samples, and extraction of their statistical features for noise type identification. The isolation of representative noise samples is achieved using simple image filters and noise identification is performed using a few statistical and/or histogram based features. The method seems to be capable of accurately classifying the types of noise studied.

I. INTRODUCTION

Over the past two decades, a wide variety of image de-noising algorithms have appeared in the literature. In general, they perform quite well, but almost all of them are geared toward reduction or removal of a specific type of noise, such as, Gaussian white noise [1]-[3], non-Gaussian or Gaussian white noise [4]-[6], speckle noise [7]-[9] and impulsive (or, salt-and-pepper) noise [3],[9],[10]. The underlying techniques include Kalman filter, Wiener filter, various wavelet de-noising schemes, median filter, order statistic filters [10] and other methods [11]. Although these techniques are very useful for applications where manual image de-noising is acceptable, they fall short of their goals in many other applications that call for automated image restoration.

In view of above, automated techniques for identification of image noise are of considerable interest, because once the type of noise is identified from the given image, an appropriate algorithm can then be used to de-noise it. Only a few researchers have addressed this issue to date [12]-[14]. However, the proposed algorithms are pretty complicated, because their main goal is to estimate the statistical parameters of the noise. For instance, the authors of [12,[13] propose a fairly complicated method based on detection of homogeneous regions, unsupervised variational classification through a multi-thresholding technique, and estimation of statistical parameters from the homogeneous regions. Finally, the identification of nature of the noise is done using a polynomial regression fit and also, the the statistical parameters of the noise are estimated from the homogeneous regions.

This paper proposes a simple technique for identification of the type of noise present in an image, which is desirable in many applications that call for superior de-noising performance. It may be pointed out that de-noising often constitutes a first step that is followed by other automated image processing operations that analyze the image and extract useful features from it. The performance of the follow-up operation is often adversely affected due to poor de-noising results obtained in the first step. Since poor de-noising often results from poor noise identification, a better noise identification technique is always preferred.

II. BASIC PRINCIPLE OF THE PROPOSED NOISE IDENTIFICATION METHOD

In principle, the noise identification method proposed here consists of three key steps:
Step 1. Extract some representative noise samples from the given noisy image,
Step 2. Estimate some of their statistical features, and
Step 3. Use a simple pattern classifier to identify the type of noise.
The above steps can be summarized as follows.

Assume the original M×N image $y(i,j)$ is contaminated by either additive or multiplicative type of noise, $\omega(i,j)$. Thus the observed image $f(i,j)$ can be modeled by either equation (1) for additive noise, or equation (2) for multiplicative noise:

$$f(i,j) = y(i,j) + \omega(i,j), \quad 1 \le i \le M, 1 \le \mathrm{j} \le \mathrm{N} \tag{1}$$

$$f(i,j) = y(i,j)\omega(i,j), \quad 1 \le i \le M, 1 \le \mathrm{j} \le \mathrm{N} \tag{2}$$

To start with we assume that the type of noise is unknown, but it belongs to one of N known classes. For each type of noise, we choose a simple linear or nonlinear spatial filter operator capable of removing most of the noise of this type from the image. Suppose $H_k(i,j)$, $1 \le k \le N$, denote these filter operators.

To extract some noise samples, first process the image through each filter operator to obtain:

$$g_k(i,j) = H_k(i,j) * f(i,j),\ 1 \le k \le N, \tag{3}$$

where * denotes the associated spatial filtering operation. Next, subtract each processed image, $g_k(i,j)$, $1 \le k \le N$, from f(i,j) to extract representative noise samples, $w_k(i,j)$, corresponding to each type of noise:

$$w_k(i,j) = f(i,j) - g_k(i,j),\ 1 \le k \le N. \tag{4}$$

Next, estimate some simple statistical and/or histogram based features from $w_k(i,j)$, $1 \le k \le N$, and then classify the noise into one of N known classes using similarity measures, S_k, $1 \le k \le N$, where S_k denotes the similarity of the features measured with respest to the kth type of noise.

The above method is also useful to investigate the presence of two or more types of noise present in a noisy image. To do so, find the probability that the noise belongs to a specific type by:

$$p(k) = S_k \Big/ \sum_{n=1}^{N} S_n \tag{5}$$

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III. IMPLEMENTATION AND INVESTIGATION

In this paper, we consider four different types of commonly occurring image noise, namely, uniform white, Gaussian white, speckle, and salt-and-pepper noise. Among these four types, speckle noise is of multiplicative type, whereas the other three are additive in nature. The filters selected for the above four types of noise are: Wiener filter [3] for uniform or Gaussian white noise, homomorphic filter [9] for speckle noise, and median filter [3],[9] for salt-and-pepper noise. Also, the statistical features studied here include "kurtosis" and "skewness". Table 1 lists the probability density functions, "kurtosis" and "skewness" values, and the selected filters for the four types of noise.

From Table 1, we can see that different type of noise has different kurtosis or skewness values and those differences can be used to identify the noise type.

Table 1. PDF, Kurtosis, Skewness, and filters selected for four types of noise

	Probability Density Function	Kurtosis	Skewness	Selected Filter
Non-Gaussian White	$f(x)=\begin{cases}\frac{1}{b-a}, & a\le x\le b\\ 0, & \text{otherwise}\end{cases}$	1.8	0	Wiener Filter
Gaussian White	$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	3	0	Wiener Filter
Speckle (exponential distribution)	$f(x)=\begin{cases}\frac{1}{\sigma^2}\exp(\frac{-x}{\sigma^2}), & x\ge 0\\ 0 & ,x<0\end{cases}$	9	2	Homomophic filter
Salt-and-Pepper	$f(x)=\begin{cases}p_a, \text{ for } x=\text{salt}\\ p_b, \text{ for } x=\text{pepper}\\ 0, \text{ otherwise}\end{cases}$	Large (typically greater than 20 or so, depending on the noise density)	0	Median Filter

Kurtosis is a measure of how outlier-prone a distribution is and skewness is a measure of the asymmetry of the data around the sample mean. Specifically, kurtosis and skewness are respectively 4th and 3rd order moments given by,

$$KURT = E(x-\mu)^4/\sigma^4 \tag{6}$$

$$SKEW = E(x-\mu)^3/\sigma^3 \tag{7}$$

First, the three selected filters, Wiener filter, homomophic filter, and median filter, are applied to the noisy image $f(i,j)$ to get three different estimates $\hat{y}_n(i,j)$ for the original image $y(i,j)$ as follows:

$$\hat{y}_{Wiener}(i,j) = f(i,j) * H_{Wiener}(i,j) \tag{8}$$

$$\hat{y}_{Median}(i,j) = f(i,j) * H_{Median}(i,j) \tag{9}$$

$$\hat{y}_{Homo}(i,j) = Exp[\log(f(i,j)) * H_{Wiener}(i,j)] \tag{10}$$

where, the symbol $*$ above denotes the associated spatial filtering operations. In equation (10), logarithm and exponential operations are needed to perform homomorphic filtering to remove the multiplicative noise.

Then we get three noise estimates based on the outputs from the three filters as follows:

$$\omega_{Wiener} = f(i,j) - \hat{y}_{Wiener}(i,j) \tag{11}$$

$$\omega_{Median} = f(i,j) - \hat{y}_{Median}(i,j) \tag{12}$$

$$\omega_{Homo} = f(i,j)/\hat{y}_{Homo}(i,j) \tag{13}$$

Next, the noise estimates obtained in equations (11)-(13) are used to identify the noise type. Suppose $\omega(i,j)$ is Gaussian white noise, then $\hat{y}_{Wiener}(i,j)$ should give the best estimate of $y(i,j)$ and therefore ω_{Wiener} given by equation (11) should be close to Gaussian white noise (with kurtosis close to 3 and skewness close to 0).

A simple minimum distance pattern classifier based on measuring the similarity of Kurtosis and Skewness values is used to evaluate how close ω_{Wiener} is to Gaussian or uniform white noise, ω_{Median} is to salt-and-pepper noise, and ω_{Homo} is to speckle noise. To measure these similarities, we need some expected reference values to compare with. During the filtering operations, since both the original image and the accompanying noise are processed by the above filters, the expected reference values can be obtained by filtering the appropriate noise sequences and evaluating the Kurtosis and Skewness of the filter outputs. For instance, to obtain the expected reference values corresponding to uniform white noise, we do the following:

- First generate training sequences of uniform white noise.
- Then, filter each noise sequence through the Wiener filter.
- Then estimate the Kurtosis and Skewness of each filtered noise sequence and compute their average to yield the reference values of Kurtosis and Skewness for the uniform white noise.

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Based on our training data sets, the following simplified rule was developed for the pattern classifier.

If $KURT_{Median} \le T$:

$$S_{UW} = 0.5 * \left[\frac{\min(KURT_{Wiener}, KURT_{UW_Wiener})}{\max(KURT_{Wiener}, KURT_{UW_Wiener})} + \frac{\min(SKEW_{Wiener}, SKEW_{UW_Wiener})}{\max(SLEW_{Wiener}, SKEW_{UW_Wiener})} \right] \quad (14)$$

where $KURT_{GW_Wiener}$ = 3.48, $SKEW_{GW_Wiener}$ = 0.04 (which are based on training data).

$$S_{GW} = 0.5 * \left[\frac{\min(KURT_{Wiener}, KURT_{GW_Wiener})}{\max(KURT_{Wiener}, KURT_{GW_Wiener})} + \frac{\min(SKEW_{Wiener}, SKEW_{GW_Wiener})}{\max(SLEW_{Wiener}, SKEW_{GW_Wiener})} \right] \quad (15)$$

where $KURT_{GW_Wiener}$ = 4.38, $SKEW_{GW_Wiener}$ = 0.04 (which are based on training data).

$$S_{Speckle} = 0.5 * \left[\frac{\min(KURT_{Homo}, KURT_{Speckle_Homo})}{\max(KURT_{Homo}, KURT_{Speckle_Homo})} + \frac{\min(SKEW_{Homo}, SKEW_{Speckle_Homo})}{\max(SLEW_{Homo}, SKEW_{Speckle_Homo})} \right] \quad (16)$$

where $KURT_{Speckle_Homo}$ = 6.48, $SKEW_{Speckle_Homo}$ = 1.45 (which are based on training data).

$$S_{S\&P} = 0$$

else if $KURT_{Median} > T$:

$$S_{GW} = 0,\ S_{UW} = 0,\ S_{Speckle} = 0$$

$$S_{S\&P} = 0.5 * \left[\frac{\min(KURT_{Median}, KURT_{S\&P_Median})}{\max(KURT_{Median}, KURT_{S\&P_Median})} + \frac{\min(SKEW_{Median}, SKEW_{S\&P_Median})}{\max(SLEW_{Median}, SKEW_{S\&P_Median})} \right] \quad (17)$$

where $KURT_{S\&P_Median}$ = 36.5, $SKEW_{S\&P_Median}$ = 0.2 (which are based on training data).

In equation (14), $KURT_{Wiener}$ and $SKEW_{Wiener}$ denote kurtosis and skewness of ω_{Wiener}; $KURT_{GW_Wiener}$ and $SKEW_{GW_Wiener}$ denote the expected kurtosis and skewness of ω_{Wiener} if the original noise $\omega(i,j)$ is actually Gaussian white noise. The ratio between real kurtosis $KURT_{Wiener}$ and the expected kurtosis $KURT_{GW_Wiener}$, as well as the ratio between real skewness $SKEW_{Wiener}$ and the expected skewness $SKEW_{GW_Wiener}$, are used as similarity measures. Minimum and maximum operations are used to assure that the maximum similarity value is 1. The similarities of kurtosis and skewness are equally weighted to get S_{GW}, the final measure of similarity between ω_{Wiener} and actual Gaussian white noise. It should be pointed out that because the Wiener filter used here doesn't completely filter out Gaussian or uniform white noise, the expected kurtosis $KURT_{GW_Wiener}$ and skewness $SKEW_{GW_Wiener}$ are usually not equal to their ideal values, 3 and 0, shown in Table 1. These expected values are obtained by using Gaussian noise sequences as training data. We have used $KURT_{GW_Wiener}$ = 4.38, $SKEW_{GW_Wiener}$ = 0.04 based on our actual training data. A similar method is used to obtain the expected kurtosis and skewness values for speckle noise and salt-and-pepper noise. Finally, based on experimental data, the value of T was chosen to be 20 in this set of experiments.

IV. EXPERIMENTAL RESULTS

All experiments were carried out using Matlab simulations. Matlab function "imnoise" is used to generate Gaussian white noise and salt-and-pepper noise. Also, "rand" function is used to create uniform white noise. Lastly, for speckle noise, we have used the noise model described in [9].

We have conducted simulations on many different image sequences. Figures (1a)-(1d) represent one of the noisy image sequences as an example. With four different noise inputs, the calculated kurtosis, skewness, their similarities, and identified noise types are shown in Table 2. For example, given uniform white noise input, the kurtosis, skewness from ω_{Wiener} matches the expected kurtosis and skewness values of a typical uniform white noise better than other types of noises, which results in S_{UW} having the maximum similarity value and therefore uniform white noise is identified.

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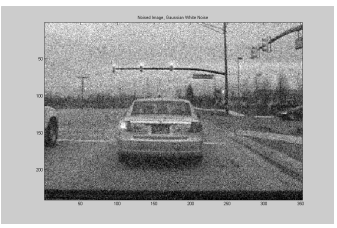
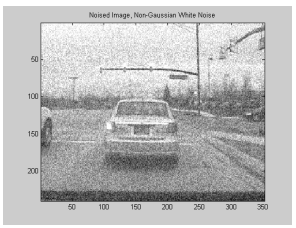
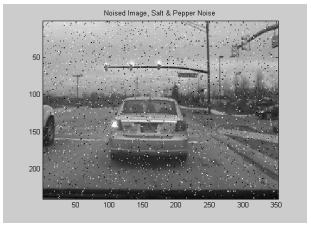
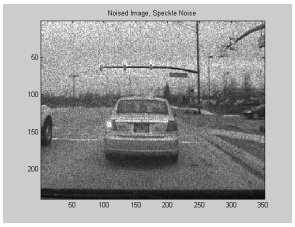
Table 2: An example of kurtosis, skewness, similarity, and identified noise types

Noise Input	K_{Wiener}	K_{Homo}	K_{Median}	SK_{Wiener}	SK_{Homo}	SK_{Median}	S_{UW}	S_{GW}	$S_{Speckle}$	$S_{S\&P}$	Noise Type Identified
UW	**3.48**	2.94	3.16	**0.05**	0.14	0.03	**0.88**	0.78	0.28	0	**1**
GW	**4.40**	80.23	4.15	**0.08**	7.11	-0.0219	0.64	**0.74**	0.14	0	**2**
Speckle	19.41	**6.39**	5.72	-2.83	**1.44**	-0.93	0.1	0.12	**0.99**	0	**3**
Salt & Pepper	49.43	105.78	**38.68**	-.02	9.52	**0.53**	0	0	0	**0.66**	**4**









Figures (1a)-(1d). Clockwise from left (Gaussian white noise, non-Gaussian white noise, impulsive noise, and speckle noise

Tests were also conducted successfully on images contaminated with mixed types of noise. For instance, for an image contaminated with both Gaussian white noise and speckle noise, the probabilities obtained were as follows:

$$p(k) = \begin{bmatrix} 0.2352 & 0.2902 & 0.4745 & 0 \end{bmatrix},$$

where k = 1, 2, 3, 4 correspond to uniform white, Gaussian white, speckle, and salt-and-pepper noise, respectively.

V. CONCLUSION

A simple technique for identifying the type of noise present in a noisy image is proposed in this paper. The proposed technique is quite general in nature and can be used with a variety of de-noising filters. The results of simulation studies seem to indicate that the method is capable of accurately determining the type of noise.

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